WORKED SOLUTION

STEP II 2000, Question 2

Pure • Repeated Roots of Polynomials

STEP II 2000, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s2-q2

The key idea

A repeated factor survives differentiation, but loses one power each time. So a fourth-power factor $(x-a)^4$ forces p and its first three derivatives all to vanish at a. For the sextic, the third derivative is only a cubic, and its roots give a very short list of candidates for a. We test those, then choose k last, since k disappears as soon as we differentiate.

Solution

The first result

Suppose $(x-a)^2$ is a factor, so $\mathrm{p}(x) = (x-a)^2\,\mathrm{q}(x)$ for some polynomial q. By the product rule,

$$\mathrm{p}'(x) = 2(x-a)\,\mathrm{q}(x) + (x-a)^2\,\mathrm{q}'(x) = (x-a)[\,2\mathrm{q}(x) + (x-a)\mathrm{q}'(x)\,].$$

Putting $x = a$ gives $\mathrm{p}'(a) = 0$, as required.

The corresponding result

The same calculation works for any power. If $\mathrm{f}(x) = (x-a)^m\,\mathrm{g}(x)$ with $m \geq 1$ and g a polynomial, then

$$\mathrm{f}'(x) = (x-a)^{m-1}[\,m\,\mathrm{g}(x) + (x-a)\,\mathrm{g}'(x)\,],$$

and the bracket is again a polynomial. So *if* $(x-a)^m$ *divides* f, *then* $(x-a)^{m-1}$ *divides* f′.

Now let $(x-a)^4$ be a factor of $\mathrm{p}(x)$. Applying this fact three times,

$$(x-a)^3 \text{ divides } \mathrm{p}'(x), \qquad (x-a)^2 \text{ divides } \mathrm{p}''(x), \qquad (x-a) \text{ divides } \mathrm{p}'''(x).$$

Together with $(x-a)^4$ dividing $\mathrm{p}(x)$ itself, setting $x = a$ gives

$$\mathrm{p}(a) = \mathrm{p}'(a) = \mathrm{p}''(a) = \mathrm{p}'''(a) = 0.$$

Finding k

Let $\mathrm{p}(x) = x^6 + 4x^5 - 5x^4 - 40x^3 - 40x^2 + 32x + k$. Differentiating three times,

$$\begin{aligned}
\mathrm{p}'(x) &= 6x^5 + 20x^4 - 20x^3 - 120x^2 - 80x + 32, \\
\mathrm{p}''(x) &= 30x^4 + 80x^3 - 60x^2 - 240x - 80 = 10\,(3x^4 + 8x^3 - 6x^2 - 24x - 8), \\
\mathrm{p}'''(x) &= 120x^3 + 240x^2 - 120x - 240 = 120\,(x^3 + 2x^2 - x - 2).
\end{aligned}$$

Ask yourself Which of the four conditions should we use first?

Use the one that is easiest to solve. The constant k has disappeared from p''', and p''' is only a cubic, which factorises by grouping:

$$x^3 + 2x^2 - x - 2 = x^2(x+2) - (x+2) = (x+2)(x-1)(x+1).$$

So by the corresponding result, a must be -2, 1 or -1. We test each in p'':

$$\begin{aligned}
\tfrac{1}{10}\mathrm{p}''(1) &= 3 + 8 - 6 - 24 - 8 = -27 \neq 0, \\
\tfrac{1}{10}\mathrm{p}''(-1) &= 3 - 8 - 6 + 24 - 8 = 5 \neq 0, \\
\tfrac{1}{10}\mathrm{p}''(-2) &= 48 - 64 - 24 + 48 - 8 = 0.
\end{aligned}$$

Only $a = -2$ survives. It also passes the p' test:

$$\mathrm{p}'(-2) = -192 + 320 + 160 - 480 + 160 + 32 = 0.$$

Finally, $\mathrm{p}(-2) = 0$ fixes k:

$$\mathrm{p}(-2) = 64 - 128 - 80 + 320 - 160 - 64 + k = k - 48 = 0, \qquad \text{so} \qquad \boxed{k = 48}.$$

Ask yourself We have only used *necessary* conditions. How do we know that $(x+2)^4$ really is a factor when $k = 48$?

The four conditions $\mathrm{p}(a) = \mathrm{p}'(a) = \mathrm{p}''(a) = \mathrm{p}'''(a) = 0$ do in fact imply that $(x-a)^4$ is a factor (by Taylor's theorem for polynomials). The quickest confirmation, though, is to multiply out. Since $(x+2)^4 = x^4 + 8x^3 + 24x^2 + 32x + 16$, the quotient must be a monic quadratic with constant term $48/16 = 3$. Matching the x^5 coefficient gives $x^2 - 4x + 3$, and indeed

$$(x+2)^4(x^2 - 4x + 3) = x^6 + 4x^5 - 5x^4 - 40x^3 - 40x^2 + 32x + 48.$$

So, with $k = 48$,

$$x^6 + 4x^5 - 5x^4 - 40x^3 - 40x^2 + 32x + 48 = (x+2)^4(x-1)(x-3),$$

and the factor has the required form, with $a = -2$.

Checks

- Expand $(x+2)^4(x-1)(x-3)$ in full: every coefficient matches the given polynomial, and the constant term is $16 \times 3 = 48$.
- Test the sum of the coefficients: $\mathrm{p}(1) = 1+4-5-40-40+32+48 = 0$, as it must be since $(x-1)$ is a factor.
- A computer confirmed the general result on random polynomials $(x-a)^4\mathrm{q}(x)$, found the roots $-2, \pm 1$ of p''' with p'' vanishing only at -2, and gave $k = 48$.

Common slips

- Starting with $\mathrm{p}(a) = 0$ or $\mathrm{p}'(a) = 0$. These involve a quintic or k itself, and lead nowhere quickly. Start from the lowest-degree derivative.
- Stopping at a root of p'''. The candidates $a = \pm 1$ are roots of p''', but they are not roots of p''.
- Only stating the corresponding result as "$\mathrm{p}'''(a) = 0$". Make it the full set $\mathrm{p}(a) = \mathrm{p}'(a) = \mathrm{p}''(a) = \mathrm{p}'''(a) = 0$, and prove it.

Beyond STEP

The converse holds too: $(x-a)^m$ divides p exactly when p and its first $m-1$ derivatives vanish at a. This is because a polynomial equals its own Taylor expansion about a. In a first-year algebra course, the same idea defines the *multiplicity* of a root. It also gives a test for repeated roots: compute $\gcd(\mathrm{p}, \mathrm{p}')$ with Euclid's algorithm. Over fields such as the integers modulo a prime, the formal derivative behaves in surprising ways. Allenby's *Rings, Fields and Groups* develops this.

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